Low Cost Driver Gaze Classification

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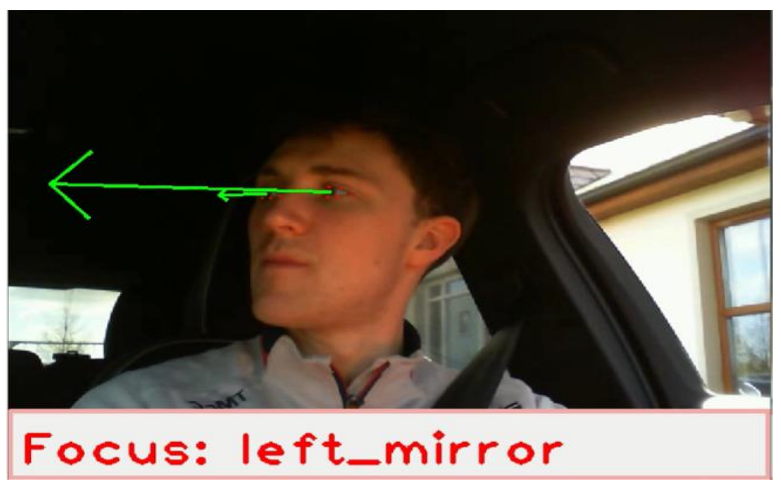
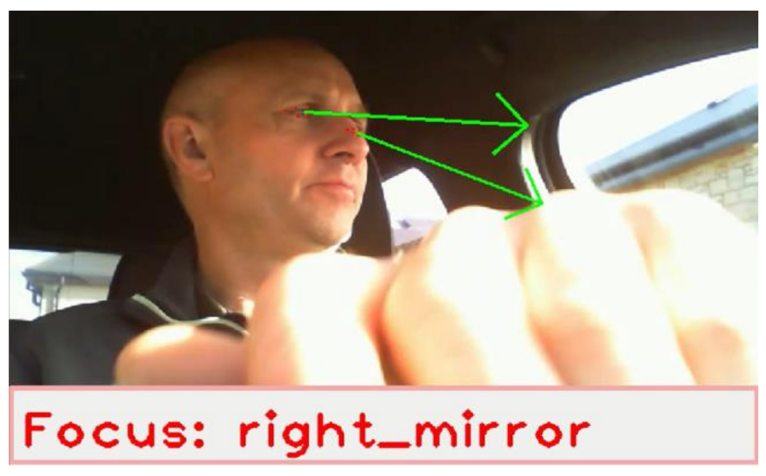




Overview

This project aims to classify drivers' gaze, focusing on key regions displaying driver awareness. This system leverages MediaPipe AI edge technology for facial and iris landmarking, extending Driver-State-Detection open source software to establish an efficient and robust system for gaze classification at a low cost.

Calibration

Calibration involves collecting the averaged gaze data associated with each point of driver focus. To ensure natural behaviours from participants, eye closures are removed, averages filter outliers and large standard deviations are highlighted as failed calibration. Audio instructions mimic driving scenarios to guide users.

Feasibility Testing

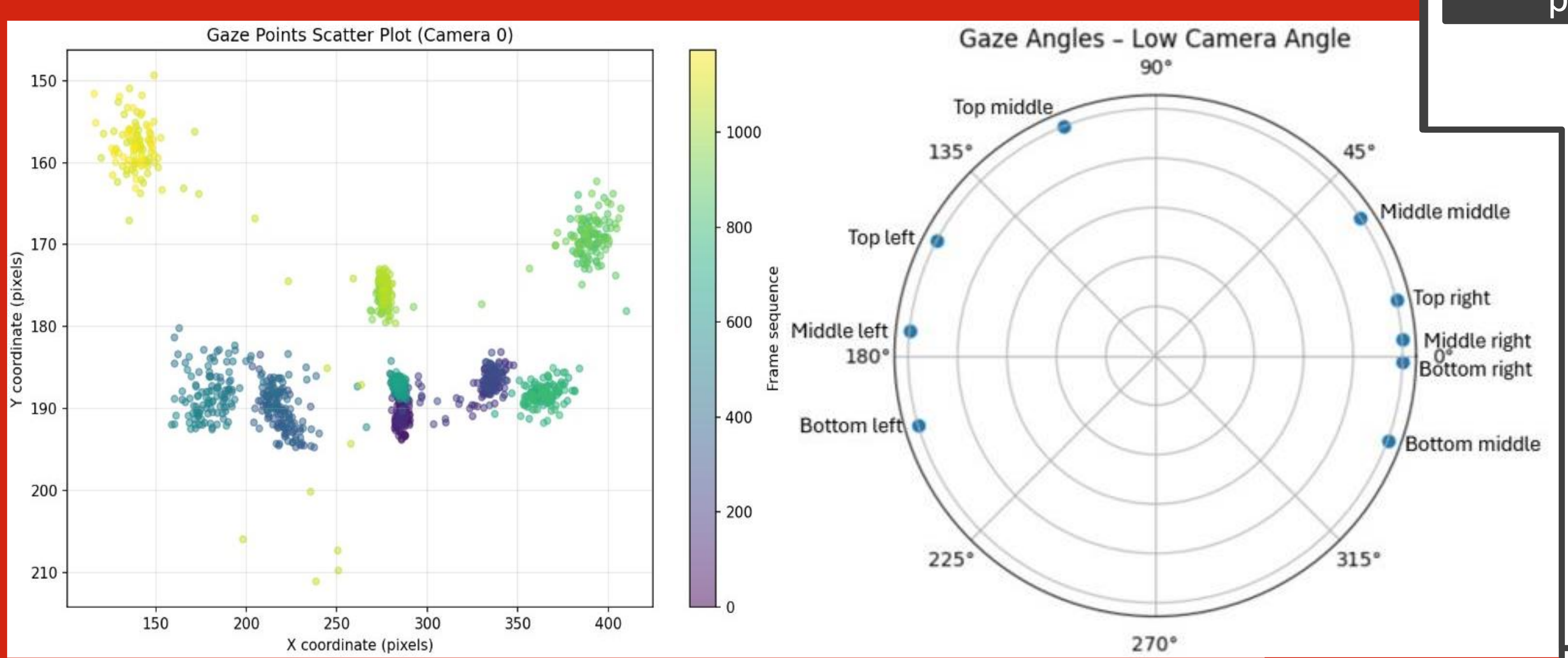
Testing showed the gaze vector can classify driver focus effectively. Achieving high accuracy with an RGB camera at an angle of elevation. Testing also identifying limitations of vertical gaze.

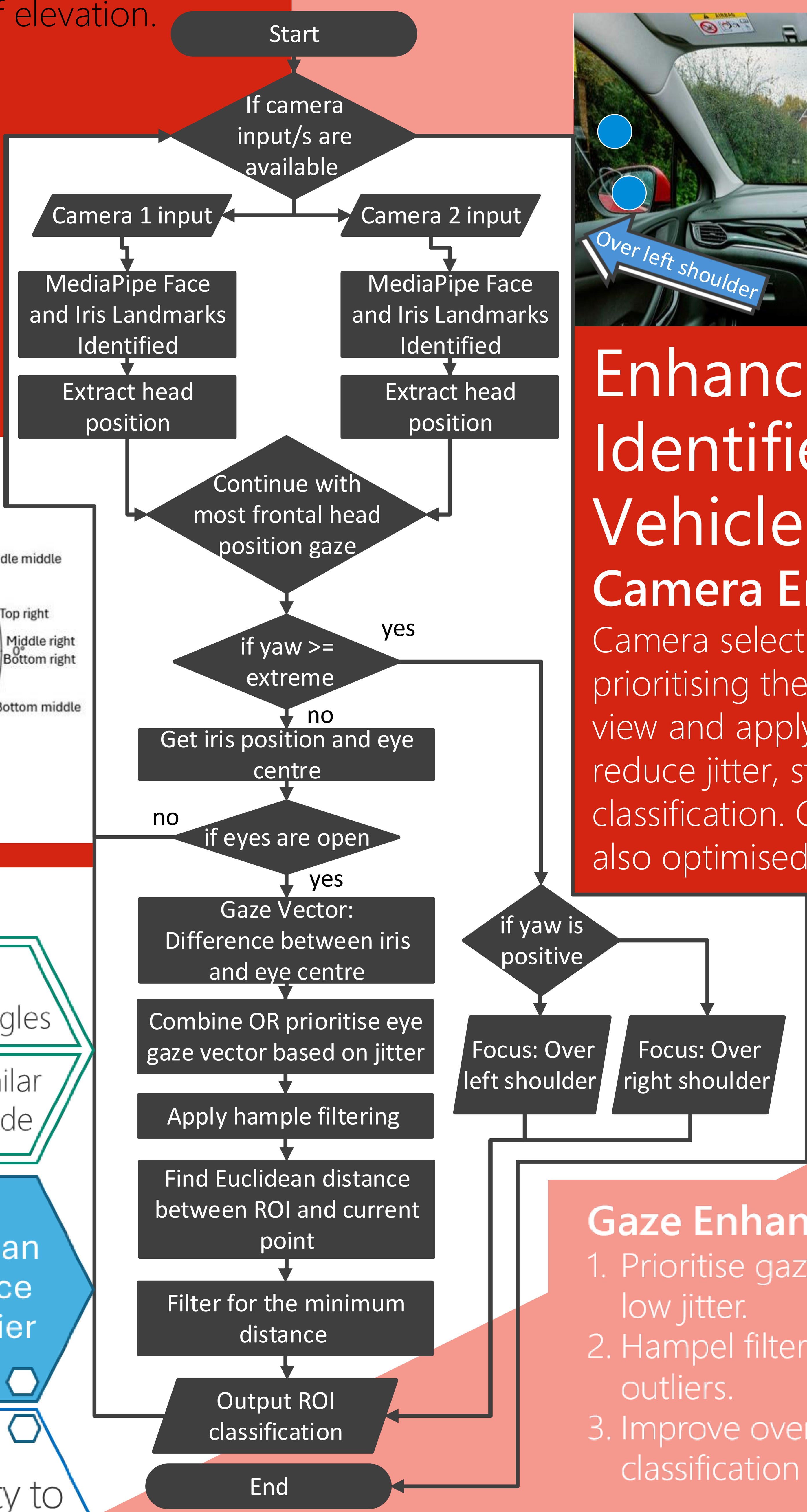
Microsoft LifeCam HD-3000

Readily available at a low cost.
Flexible mounting options in vehicle.
Compatible with MediaPipe's training data.
Real time capabilities.

Gaze distinguishability

Gaze was analysed by point and angle on a 9-point grid.

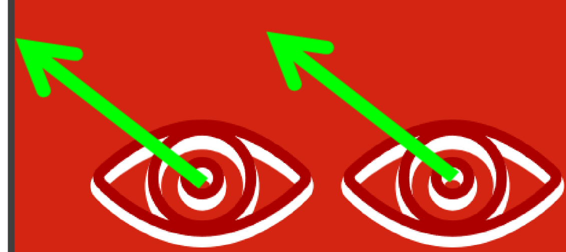




Enhancements Identified Through Vehicle Testing

Camera Enhancements

Camera selection was optimised by prioritising the most frontal head view and applying thresholding to reduce jitter, stabilising gaze classification. Camera positioning is also optimised for gaze regions.

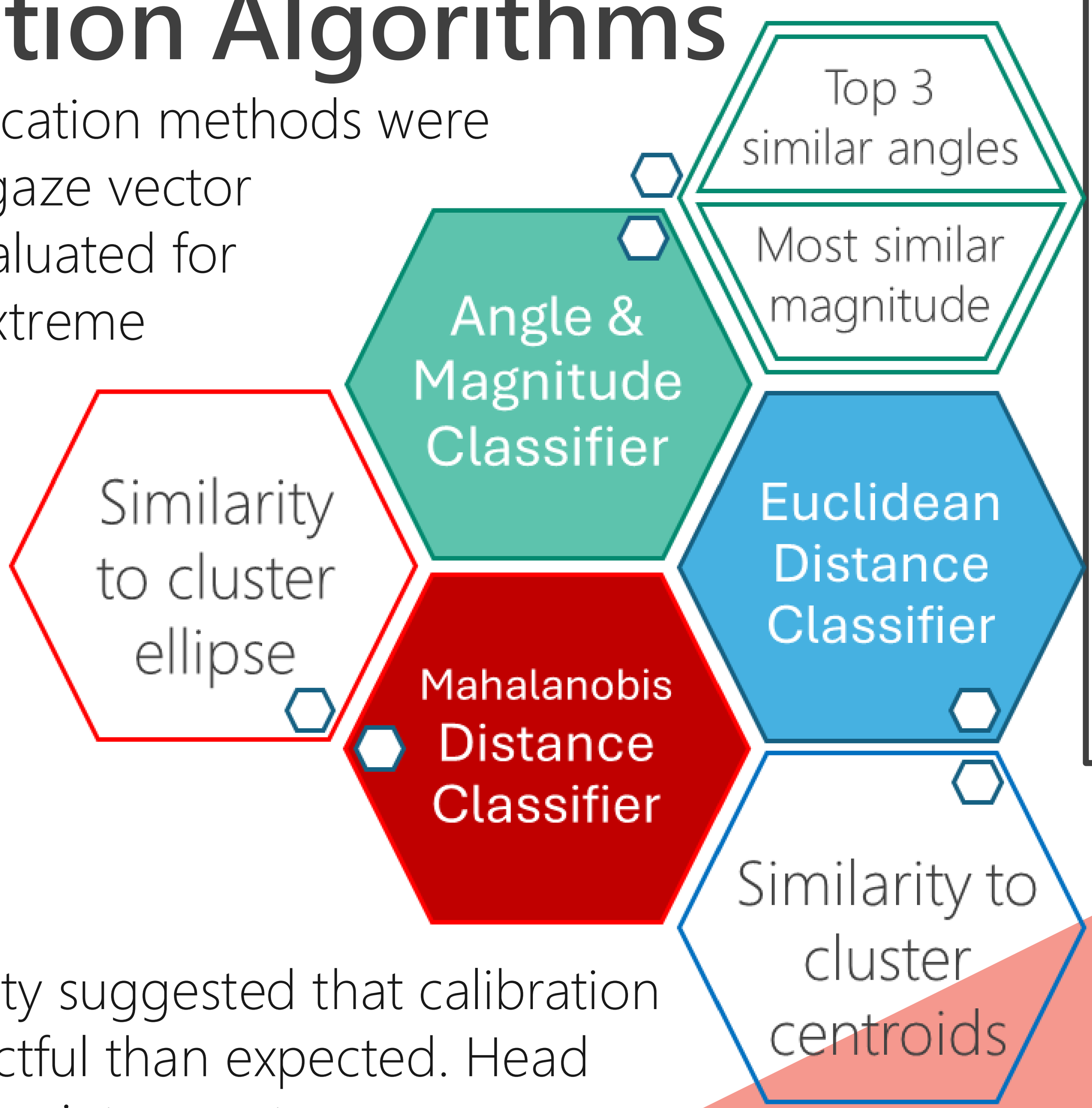


Gaze Enhancements

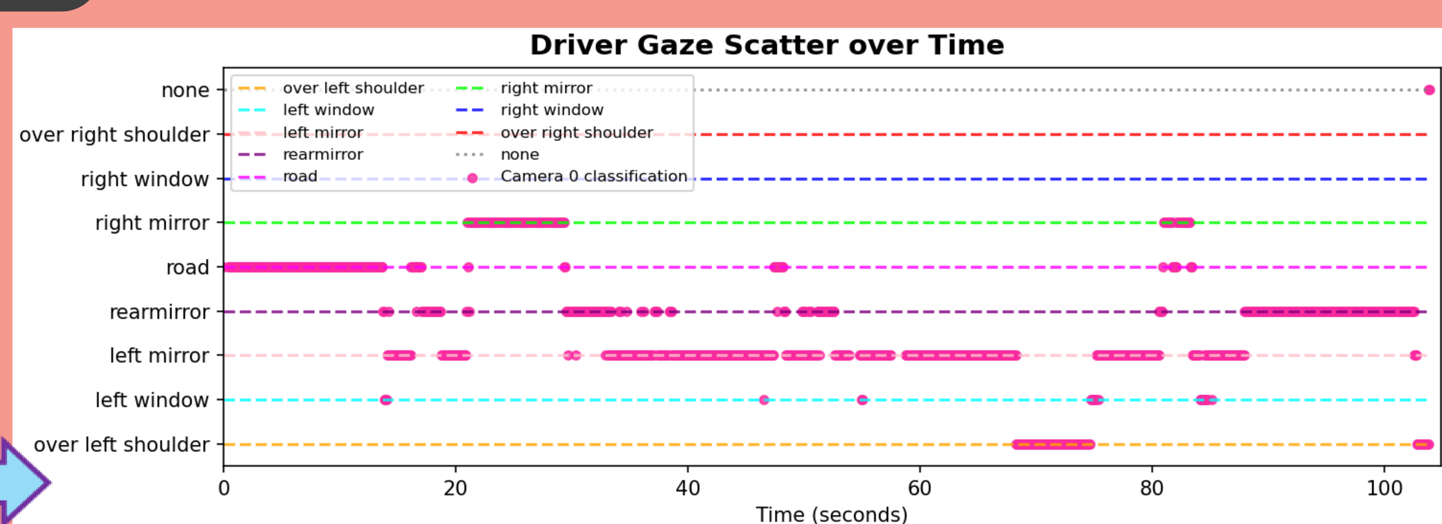
1. Prioritise gaze vector based on low jitter.
2. Hampel filtering to remove outliers.
3. Improve over the shoulder classification with yaw value.

Classification Algorithms

Three gaze classification methods were developed using gaze vector behaviour and evaluated for accuracy across extreme head and eye movements, participant variability, and calibration drift.



Reverse Around the Corner Gaze



Results

Observed variability suggested that calibration drift was less impactful than expected. Head postion will be taken into greater consideration and overall, Euclidean based classification will be used.

Gaze Accuracy Comparison	Head	Eye	Varying positions	Varying participants	Overall
Angle & Magnitude	49	64	57	55	56
Euclidean	54	65	66	62	62
Mahalanobis	34	48	-	48	43

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